BioMatch – Developer brief

Scoring KPIs, weights, formulas and system structure · v2.0 · 24 Oct 2026 · Reference implementation: `question-bank.json` + `scoring.js` (plain JS, no dependencies)

Scope. Biotech therapeutics marketplace. Startups fill a non-confidential Startup CV → **Readiness Score** (0–100, listed at ≥ 60). Listed startups and verified VCs get a reciprocal **Matching Score** (0–100). No confidential data is stored; CDA, data room and diligence happen off-platform after both sides accept an intro.

1 · System structure

Flow

1. **Onboarding** – TRL answer → band D1–D4; modalities → modules; company type → platform module
2. **Startup CV** – profile fields (18, unscored) + scored questions of the template
3. **Verification** – public IDs checked against registries (DOI, patent no., NCT, grant ID)
4. **Readiness** – score + 5 gates + 10 consistency checks → level
5. **VC profile** – thesis, must-haves, fund capacity (verified VCs only)
6. **Matching** – hard filters → two fits → geometric mean → explanation
7. **Intro** – request (text, ≤ 500 chars) → accept / pass with reason → contacts released

Core entities

Entity	Key fields
Startup	profile P01–P18, band, modules, answers{}, deck, confirmedWarnings[], updatedAt
Answer	questionId, option | options[] | items[] | value, proof{value, type, verified}
VC	type, rounds[], bands[], modalities[], areas[], must{}, ticket{min,max}, roles[], geography{}, valueAdd[], investmentPeriodEnd, reserveRatio, weeksToTermSheet, portfolio[], thesis
Match	startupId, vcId, score, vcFit, startupFit, conflict, failures[], strengths[], gap
IntroRequest	from, to, message, status (open / accepted / passed), passReason

2 · Template selection

Band	TRL	Evidence that exists	Typical round
D1 Discovery	1–3	Hypothesis, first proof of concept	Pre-seed, seed
D2 Lead and candidate	4	Candidate PoC and early safety in a defined model	Seed, Series A
D3 IND/CTA-enabling	5	Preclinical package, IND/IMPD in preparation	Series A
D4 Clinical	6–7	Phase 1 / Phase 2 data	Series B+

Each question has `req = [D1, D2, D3, D4]` with **R** required, **O** optional bonus, **–** hidden. Visible set = core questions + active modules (SM, BI, CT, GT, RN, PL) where `req[band] ≠ '–'`. Totals: 46 core, 19 module (+4 descriptive, unscored), 18 profile fields. Example D2 + small molecule = 47 scored questions.

3 · Readiness Score – KPI weights

Nine dimensions (KPIs). Weights per band, each column sums to 100.

ID	Dimension	D1	D2	D3	D4	Questions
TM	Team	25	20	15	15	TM01–TM07
SC	Science and evidence	30	25	20	20	SC01–SC08 (+ module)
PD	Product and plan	10	15	15	10	PD01–PD05 (+ module)
IP	IP and exclusivity	10	10	15	10	IP01–IP05
RC	Regulatory and clinical	5	5	10	15	RC01–RC05
CM	CMC	0	5	10	10	CM01–CM04 (+ module)
MK	Market	10	10	5	10	MK01–MK05
FN	Financing	5	5	5	5	FN01–FN03
EV	External validation	5	5	5	5	EV01–EV04

Modifier: if module CT or GT is active → SC −5, CM +5 (all bands).

Formulas

```
points(answer) ∈ [0, 1]          // option table in question-bank.json; unanswered = 0

s_d = min(1, mean(R_d) + 0.1 × mean(O_d))   // R_d required, O_d optional answers
s_d = max(O_d)        // if d has no required question at the band; EV always

Readiness = Σ_d w_d(band) × s_d             // 0–100, 1 decimal
```

Special answer types

Question	Rule
SC02 evidence checklist	`min(1, hits / |expected(band)| + 0.05 × extras)`; expected D1 {ivpoc, te}, D2 +{ive, pkpd}, D3 +{ngtox, gtox}, D4 +{hsafe}
SC07 positioning	option points × 0.7 if fewer than 3 differentiation bullets
PD05 months to inflection	≤ 24 → 1 · 25–36 → 0.7 · > 36 → 0.4
IP05 multi-select	max of ticked options

Proof rule and evidence confidence

```
if option.proof && !answer.proof:
    points = max(points of lower options without proof) or 0

confidence = (verified + 0.5 × linked) / proofEligible   // 0.5 if none
```

Proof type	Check
doi	Crossref REST API GET `api.crossref.org/works/{doi}` – no key
trial	ClinicalTrials.gov API v2 GET `/api/v2/studies/{NCT_ID}` – no key, ~50 req/min
patent	EPO OPS `/published-data/publication/epodoc/{no}/biblio` – free OAuth (stub for demo)
grant	NIH RePORTER API v2 `/v2/projects/search`; EU grants via link
link, names	Stored as "linked" (0.5), spot-checked manually

Gates (all must pass to list)

ID	Rule
G1	Deck uploaded, non-confidential attestation ticked, scan findings resolved
G2	TM01 ≥ "One" full-time founder/executive
G3	All profile fields and required questions of the template answered
G4	No unconfirmed warnings (F-checks with severity warning)
G5	Profile updated within 6 months

Consistency checks

ID	Trigger	Severity → effect
F1	Series A at D1, or Series B/C+ at D1–D2	Warning → confirm
F2	D4 without verified trial ID	Warning → confirm
F3	IP02 = university, no agreement, at D3–D4	Critical → IP capped at 0.5, shown to VCs
F4	Human data (hsafe/heff) below D4	Warning → confirm
F5	PD05 < 6 months	Warning → confirm
F6	FN02 = under 12 months	Warning → confirm
F7	PD01 = undecided at D3–D4	Warning → confirm
F8	SC03 = single lab at ≥ D2	Open question → shown to VCs
F9	TM02 = none and TM06 = none at ≥ D2	Open question → shown to VCs
F10	RC01 = not defined at ≥ D2	Warning → confirm

Levels and feedback

Score	Level	Effect
< 60 or gate failed	Draft	Visible only to the startup
60–74.9	Listed	Visible to verified VCs; can contact VCs
75–89.9	Investor-ready	Badge, ranked higher
90–100	Top profile	Featured in weekly digest

Next best steps: for each visible question, re-score with one candidate change; gain = new − current. Quick wins: answer open question, add missing proof, add bullets. Milestones: next higher option, missing expected evidence item (skip questions with `actionable: false`). Sort by gain desc, ties → quick win first; show top 3.

4 · Matching Score

Step 1 – hard filters (any fail → score 0, pair hidden, reason logged)

- Round ∉ VC rounds · VC min ticket > startup max raise · startup needs lead and VC cannot lead
- VC must-haves not met: band, modality, area, geography, min IP, human data · exclusions (e.g. company type)
- Readiness < VC minimum (default 60) · VC type ∉ startup's wanted types · VC not accepting inbound · fund past investment period

Step 2a – VC fit (sum 100)

Component	w	f ∈ [0,1]
Therapeutic area	20	share of startup areas covered
Modality	15	share of startup modalities covered
Band	15	accepted 1 · adjacent 0.5 · else 0
Round	15	accepted 1 · adjacent 0.5 · else 0
Geography	10	country 1 · region 0.7 · global 0.6
Asset vs platform	5	match 1 · VC neutral 0.7 · mismatch 0
Thesis similarity	10	embedding cosine, rescaled (demo: `min(1, jaccard/0.25)`)
Readiness + evidence	10	`0.7×(R−60)/40 + 0.3×confidence`, clamp 0–1

Step 2b – Startup fit (sum 100)

Component	w	f ∈ [0,1]
Ticket and role	30	role OK and VC max ≥ 20% of raise mid 1 · ≥ 10% 0.6 · else 0.3
Value-add	25	share of wanted value-add offered · none wanted 0.7
Investor type	15	wanted 1 · startup open to all 0.7
Fund capacity	20	in period and reserves ≥ 40% 1 · else 0.6
Decision speed	10	weeks to term sheet ≤ months to close × 4.3 → 1 · else 0.4

Step 3 – combine

```
F_vc = Σ w_c × f_c        F_su = Σ w_c × f_c          // each 0–100

Match = round( sqrt(F_vc × F_su) − 10 × conflict )
// conflict = VC portfolio company with the same target AND indication
Label: ≥ 75 Strong · 50–74 Good · < 50 Weak (hidden by default)
Explanation: top 3 components with f ≥ 0.7 (by w×f)
Gap: component with f < 0.5 and the largest w×(1−f)
```

5 · Test values (examples.js)

Case	Expected result
Nuvaris, final (D2 + SM)	Readiness 65.2, Listed · TM 13.9 · SC 16.6 · PD 11.1 · IP 5.1 · RC 2.5 · CM 3.1 · MK 7.0 · FN 2.9 · EV 3.0
Nuvaris, draft	61.3, Draft (G3: PD02, MK04 open) · top actions: TPP up to +3.0, partnering thesis up to +2.5, regulatory pathway +2.5
Match Rheinland Bio Ventures	VC fit 89.6 · startup fit 100 · 95 Strong
Match Helvetia Life Science Fund	79.9 · 75.3 · conflict → 68 Good
Match Northstar Generalist	31.4 · 67.0 · 46 Weak
Match Atlantic Precision Partners	Filtered: round, must-have modality, readiness < 70

6 · Non-functional rules

- **Never store:** structures, sequences, constructs, formulations, unpublished data, patent drafts, FTO opinions, term sheets, cap tables, patient data.
- **Upload scan:** keywords ("confidential", "proprietary", "do not distribute", "under CDA", "trade secret") → founder confirms or replaces; regex blocks ≥ 20 nucleotide letters `[ACGTU]{20,}`, ≥ 30 amino-acid letters, SMILES-like strings, "SEQ ID NO"; strip PDF metadata, comments, speaker notes.
- **Access:** startup profiles only for verified VCs; intro messages text-only, no attachments; max 10 open intro requests per startup; contacts released after mutual accept. EU hosting; decks watermarked per viewer.